22 · Custom Directives

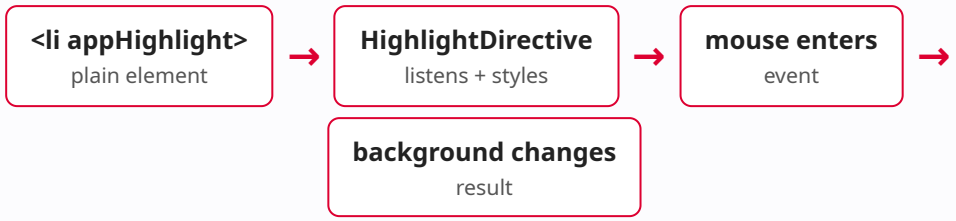
Intermediate Angular learning notes

a. Definition

A **custom directive** adds reusable behaviour to any existing element, without a template of its own. You create one with `@Directive` and a CSS-attribute **selector** such as `[appHighlight]`.

The `host` property connects the directive to the element it sits on: `'(mouseenter)'` listens to events and `'[style.color]'` binds styles. Directives can also have `input()`s and `output()`s like components.

b. Diagram



The directive attaches behaviour to a normal element

c. Syntax

```
@Directive({
  selector: '[appHighlight]',
  host: { '(mouseenter)': 'onEnter()', '[style.background-color]': 'color()' },
})
export class HighlightDirective {
  color = input('yellow', { alias: 'appHighlight' });
}

<li appHighlight="lightgreen">Idli</li>
```

d. Example Program

Programs/22_Custom_Directives/highlight.directive.ts

```
import { Component, Directive, input, output, signal } from '@angular/core';

// An attribute directive: adds hover highlighting to ANY element
@Directive({
  selector: '[appHighlight]',
  host: {
    '(mouseenter)': 'enter()',
    '(mouseleave)': 'leave()',
    // style binding on the host element
    '[style.background-color]': 'active() ? color() : null',
  },
})
export class HighlightDirective {
  color = input('yellow', { alias: 'appHighlight' }); // <li appHighlight="lightgreen">
  hoverChange = output<boolean>();                     // tell the parent about hover
```

```
  active = signal(false);

  enter() {
    this.active.set(true);
    this.hoverChange.emit(true);
  }
  leave() {
    this.active.set(false);
    this.hoverChange.emit(false);
  }
}

@Component({
  selector: 'app-root',
  imports: [HighlightDirective],
  template: `
    <ul>
      @for (item of items; track item) {
        <li appHighlight="lightgreen" (hoverChange)="onHover(item, $event)">{{ item }}</li>
      }
    </ul>
    <p>Hovered: {{ hovered() || 'nothing' }}</p>
  `,
})
export class AppComponent {
  items = ['Idli', 'Filter Coffee', 'Vada'];
  hovered = signal('');

  onHover(item: string, isOver: boolean) {
    this.hovered.set(isOver ? item : '');
  }
}
```

e. Output

Rendered in the browser (text on screen, after the UI steps below):

1. Move the mouse over **Filter Coffee**.

```
Idli
Filter Coffee
Vada
Hovered: Filter Coffee
```

f. Real-World Example (English + தமிழ்)

English

A teacher uses a yellow highlighter on any book line. The text does not change; only the look does. An attribute directive works like that highlighter on any element.

தமிழ்

பள்ளியில் ஆசிரியர் ஒரு மஞ்சள் மார்க்கர் (highlighter) பேனாவை வைத்திருக்கிறார். எந்தப் புத்தகத்தின் எந்த வரியிலும் அதைப் பயன்படுத்தி முக்கியமான இடத்தை அடையாளம் காட்டலாம். வரியின் உள்ளடக்கம் மாறுவதில்லை; அதன் தோற்றம் மட்டும் மாறுகிறது.

Custom directive அந்த மார்க்கர் பேனா: `appHighlight` என்று எந்த element மீதும் எழுதிவிட்டால் போதும், அதில் மவுஸ் வந்தால் நிறம் மாறும். ஒரு முறை எழுதி எங்கும் மறுபடி பயன்படுத்தலாம்.

g. Pros

- Reusable behaviour
- No template needed
- Composable with `hostDirectives`

h. Cons

- Can hide behaviour from template readers
- Over-engineering risk

i. Advanced / New Features

- `host` metadata object
- Directive composition API
- Signal inputs for directives

j. Common Mistakes

- Using a selector without brackets (`appHighlight` instead of `[appHighlight]`) so it matches an element name.
- Not prefixing the selector (use `app` or your own prefix) and clashing with HTML attributes.
- Manipulating `nativeElement.style` directly when a host binding would do.
- Forgetting to import the directive in the component that uses it.

k. Interview Questions

Q1. What is a directive?
A class that adds behaviour to elements. Types: components, structural and attribute directives.

Q2. How is an attribute directive created?
With `@Directive({ selector: '[name]' })`, using `host` bindings/listeners or `inject(ElementRef)`.

Q3. What does `host` do?
It binds properties, attributes, styles and events on the element that hosts the directive.

Q4. Directive vs component?
A component is a directive with a template; a directive has no template.

l. Practice Exercises

1. Create `appAutoFocus` which focuses its element when the page loads.
2. Write `appUppercase` that converts typed text in an input to capital letters.
3. Build `appTooltip="text"` that shows the text as the element's `title` attribute.